

MADEO: An LLM-Enhanced Tool for Superconducting Magnet Design & Optimization

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Abstract:

The Superconducting Magnet Division (SMD) at Brookhaven National Laboratory (BNL) has the unique capability of developing direct wind magnets, such as those that are being developed for the Electron-Ion Collider (EIC). MADEO (Modeler and Analyzer for Direct-wind Electromagnetic Optimization) is a new program that we have developed to make improvements on the legacy Fortran code that has previously been used for magnet design and development. MADEO was initially developed by hand as a small terminal-based program, after which large-language model (LLM)-assisted development was used to rapidly expand the program's capabilities. This allowed us to improve user experience by easily adding in a graphical user interface (GUI), better visualization tools, and a suite of magnet calculation tools. We were able to verify the numerical accuracy of this program by comparing results from MADEO to those of the legacy Fortran code for multiple single-parameter optimization problems in different magnet types. These comparisons produced identical results between the programs, validating the numerical accuracy of MADEO. MADEO is entirely built in Python, making it highly modular and allowing developers and physicists to integrate multiple modules natively within a single programmatic framework. It is designed to interface with modern scientific computing libraries including SciPy, C++ wrapped frameworks like Minuit, and a custom C# based geometric modeler to name a few. MADEO will allow magnet designers at the SMD at BNL to design magnets in a faster and more user-friendly way, with better visualizations of the actual magnet designs. More broadly, this work demonstrates how LLM-assisted development can accelerate the modernization of specialized scientific software from terminal based programs to user friendly GUI based modular platforms, while preserving the overall scientific functionality.

Introduction: Direct-Wind Magnets:

- Direct-wind magnets are a unique BNL technology, where superconducting wires are directly placed superconducting coils wrapped around a support structure (Fig. 1). Because they can carry very high currents, they produce strong magnetic fields, and by optimizing coil geometry, fields can be shaped for different experimental needs.
- In particle accelerators, the strength and type of magnetic field matter because they directly affect how particles move. For dipole magnets, the field strength determines how strongly the beam is bent and therefore links the accelerator's bending radius to the beam energy it can reach (Fig. 2).
- A stronger, uniform dipole field helps an accelerator reach higher energies with less complexity and energy cost. Other magnets, like Quadrupole magnets, focus the beam by pushing particles toward the center of the beamline. Sextupole magnets help keep particles with different energies properly focused.



(Fig. 1: A direct-wind magnet being wound at BNL [1])

$$F_{\text{Mag}} = ma_c \Rightarrow qvB = m \frac{v^2}{r} \Rightarrow qvB = (mv) \frac{v}{r} \Rightarrow \boxed{p = qBr}$$
$$E \approx pc \Rightarrow \boxed{E \approx (qBr)c}$$

(Fig. 2: The equation that relates the energy of a particle in a loop to the dipole field applied to it)

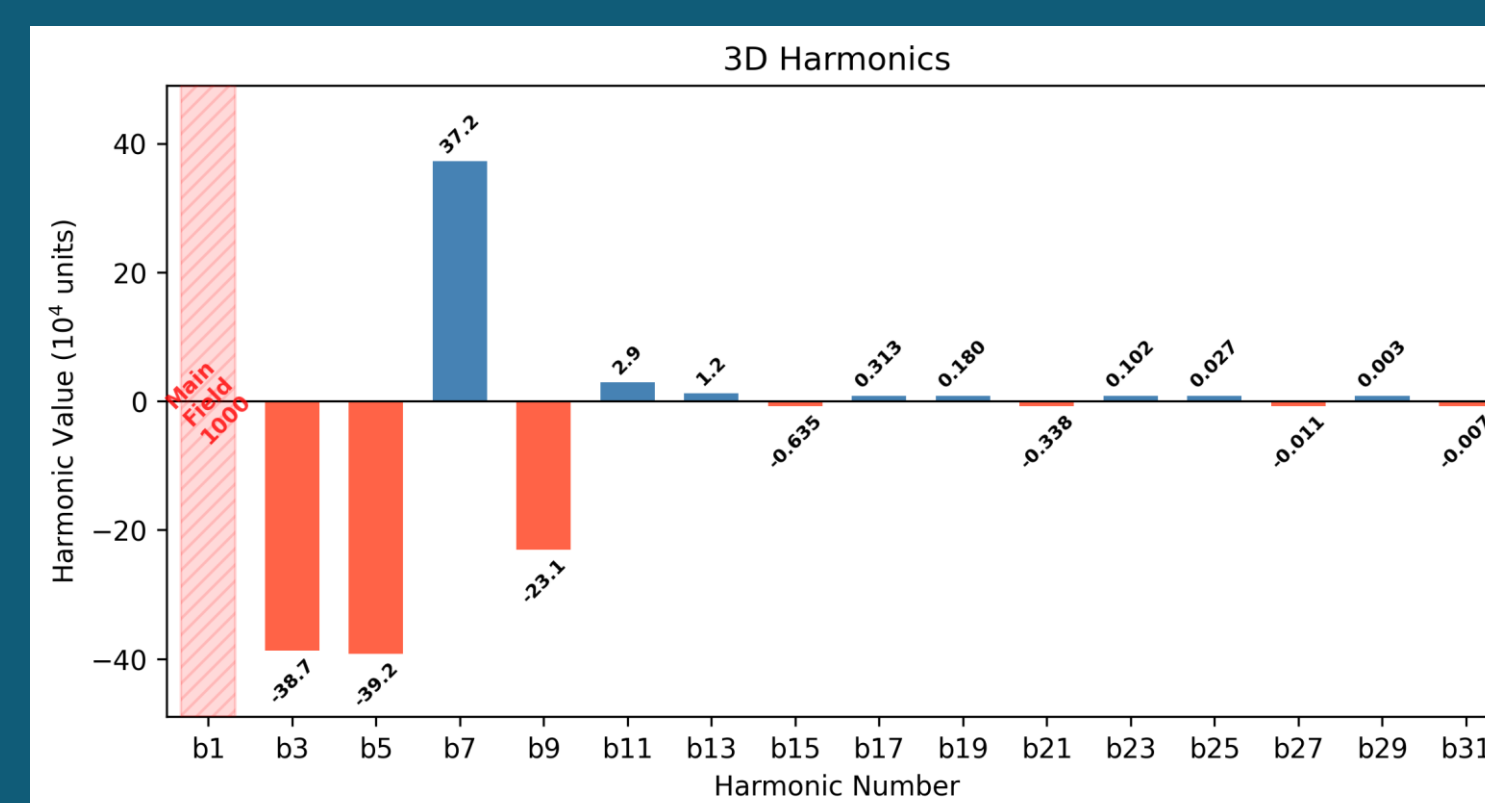
Introduction: Field Harmonics:

- Magnetic fields are composed of an infinite sum of terms called *normal harmonics* and *skew harmonics*, the mathematical definitions of which are shown in Fig. 3. Each type of magnet, or *pole*, has an associated harmonic term
- To ensure the desired accelerator field, or *main field*, is maximized, we want to minimize undesired field harmonics, so that the main field has the best quality it can have. The region through which we want to minimize these undesired harmonic terms is called the *reference radius*. This is the main radius through which the beam passes in an accelerator
- MADEO is a program that minimizes these undesired harmonics by finding the optimal spacing between conductor blocks

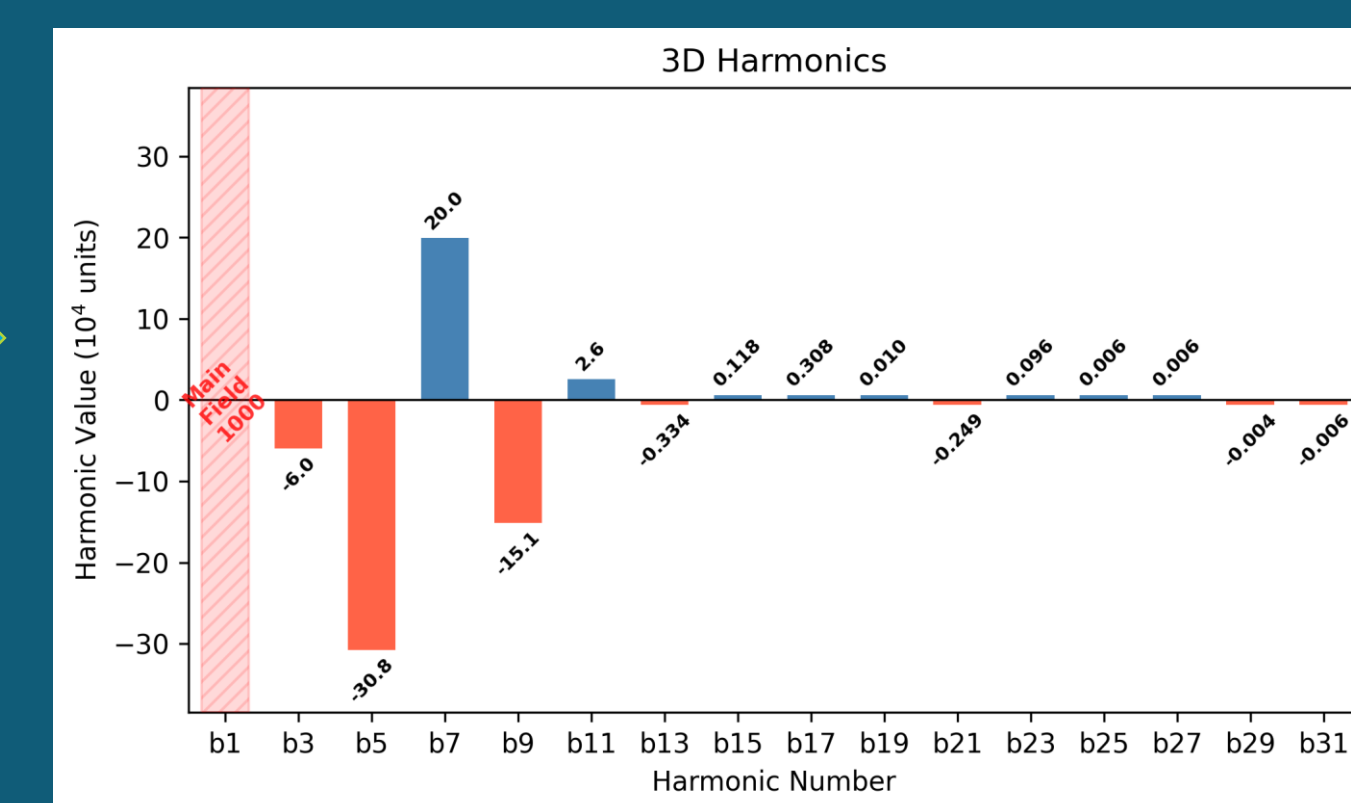
$$b_n = \frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \cos[(n+1)\phi]$$
$$b_{\text{norm}} = \frac{b_n}{b_{\text{main}}} \cdot 10^4$$
$$a_n = -\frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \sin[(n+1)\phi]$$
$$a_{\text{norm}} = \frac{a_n}{b_{\text{main}}} \cdot 10^4$$

(Fig. 3: The equations for the harmonics (b) as well as the skew harmonics (a) and their normalized form [2])

Results: Undesired Harmonic Reduction & Improved Visualization:



(Fig. 6: Harmonics pre-optimization)

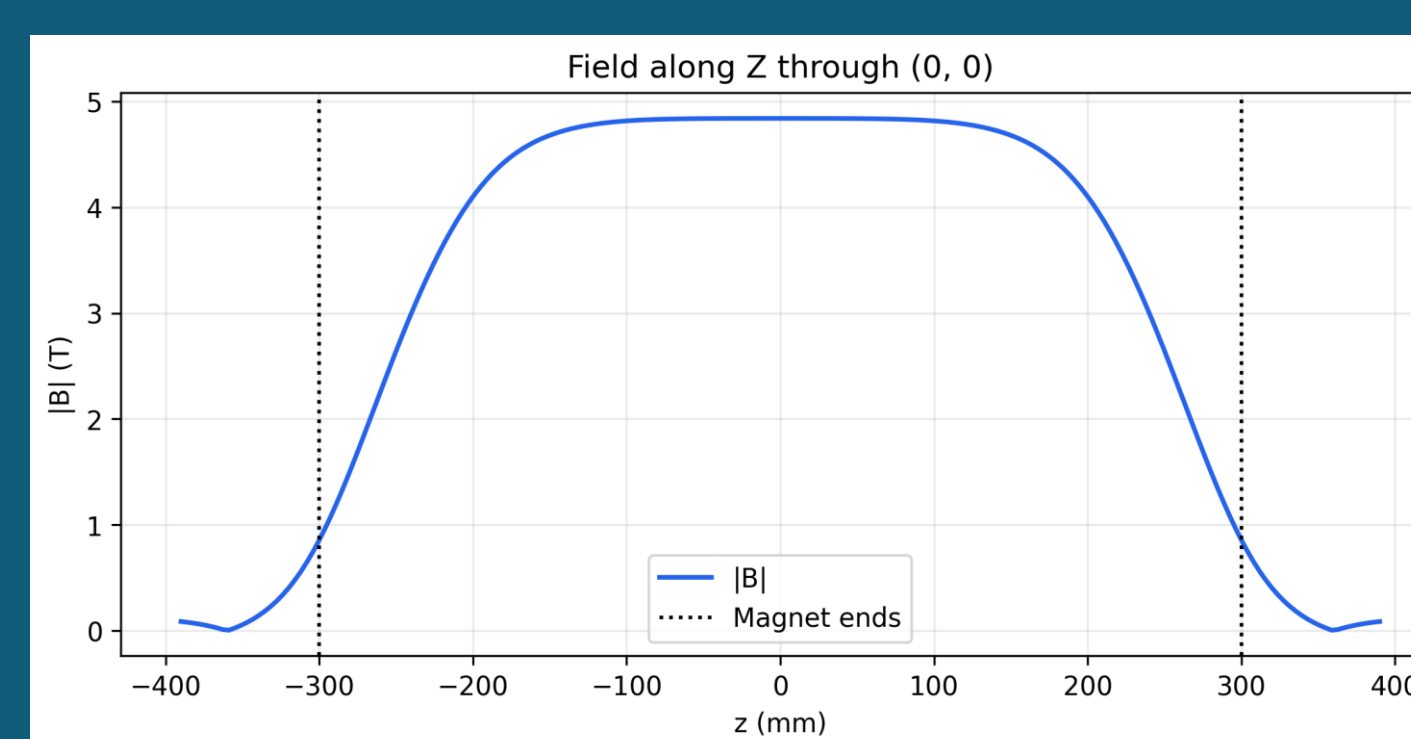


(Fig. 7: Harmonics post-optimization)

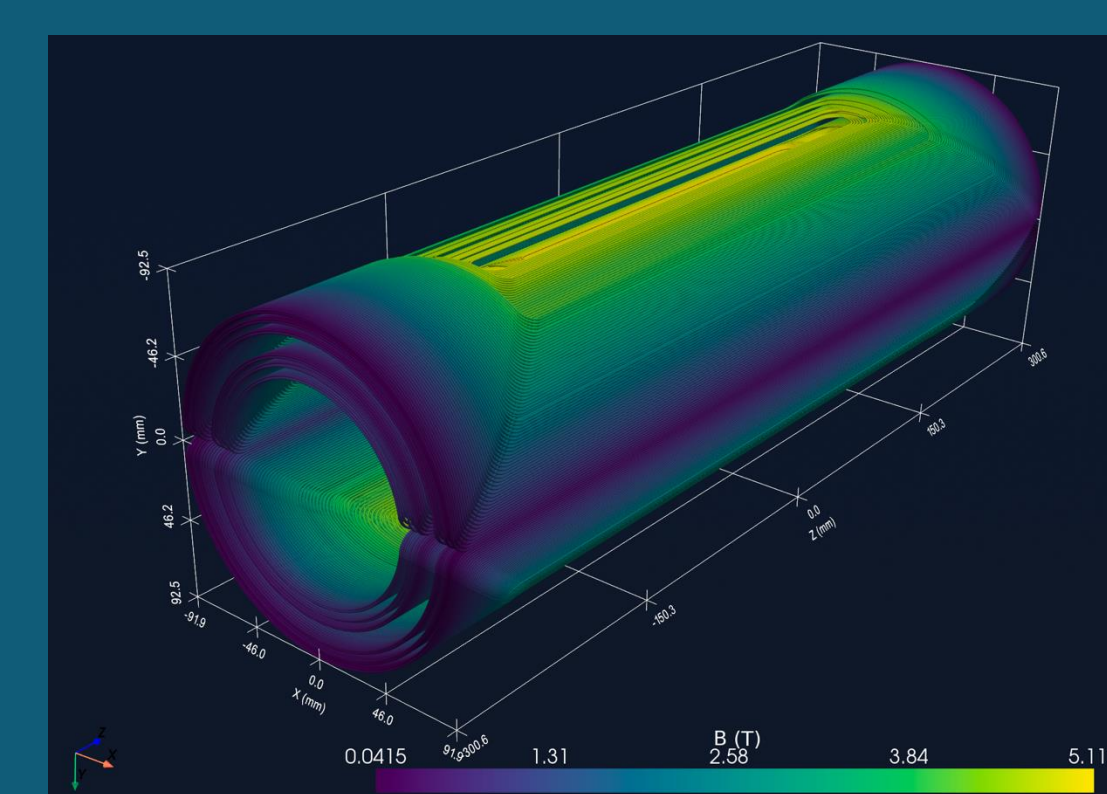
Harmonic	Weight	Pre-Optimization	Post-Optimization
b_1	0	10000.0000	10000.0000
b_3	10	-38.6835	-5.9693
b_5	10	-39.1783	-30.7688
b_7	5	37.2352	19.9527
b_9	2	-23.1072	-15.1106
b_{11}	1	2.9237	2.6041

(Table 1: Pre & post-optimization harmonic values for a 2-layer magnet example)

- In the above example, we take a 2-layer magnet and run it through the MADEO optimization algorithm
- We can see that the values produced by MADEO are indeed reduced in accordance with their individual weights
- Notably, these results also match the optimization results of the legacy Fortran code as well, confirming that MADEO is coming to the correct optimal values and geometry



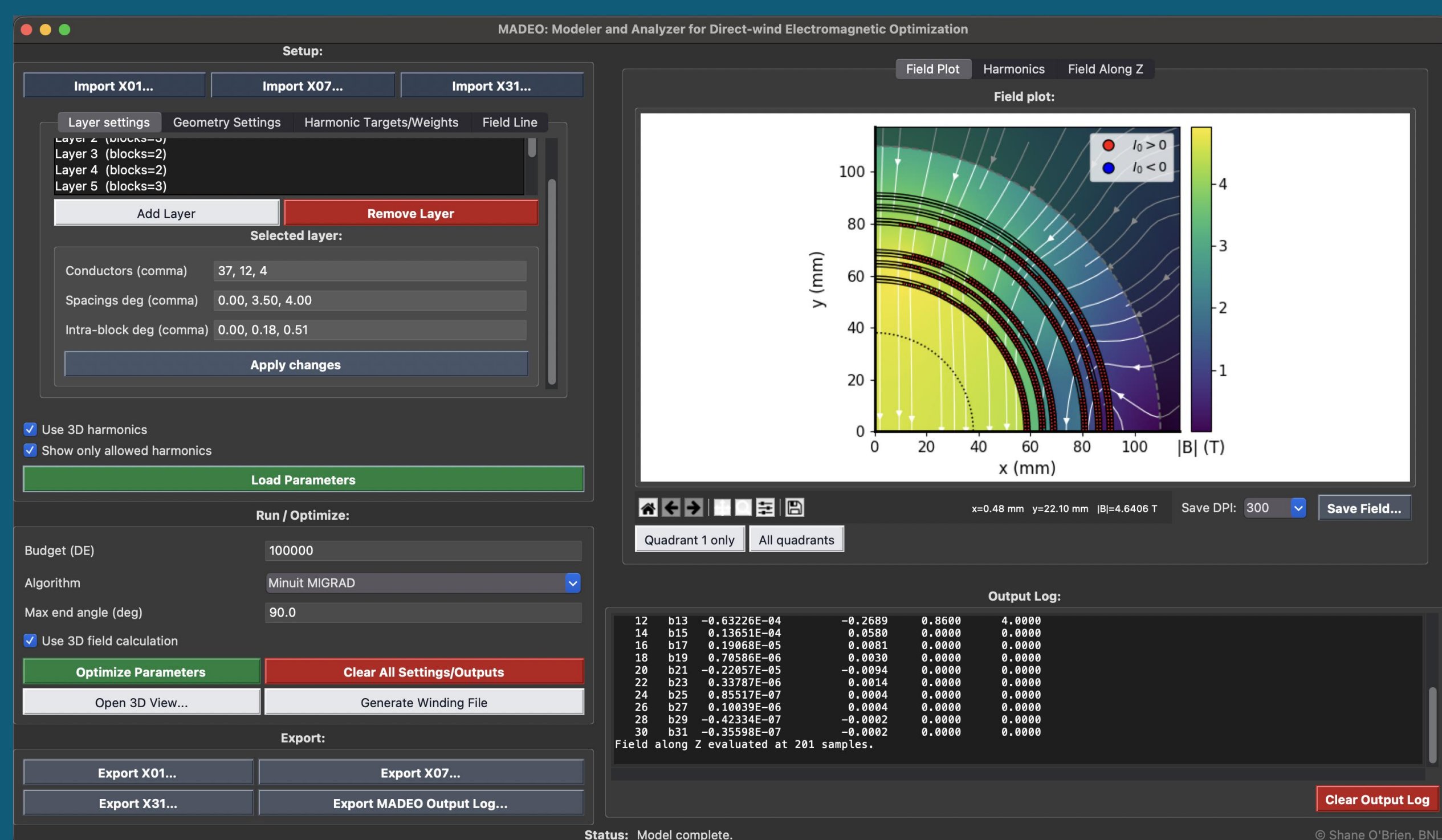
(Fig. 8: Field value through a magnet in 3D)



(Fig. 9: 3D model of a 12-layer magnet)

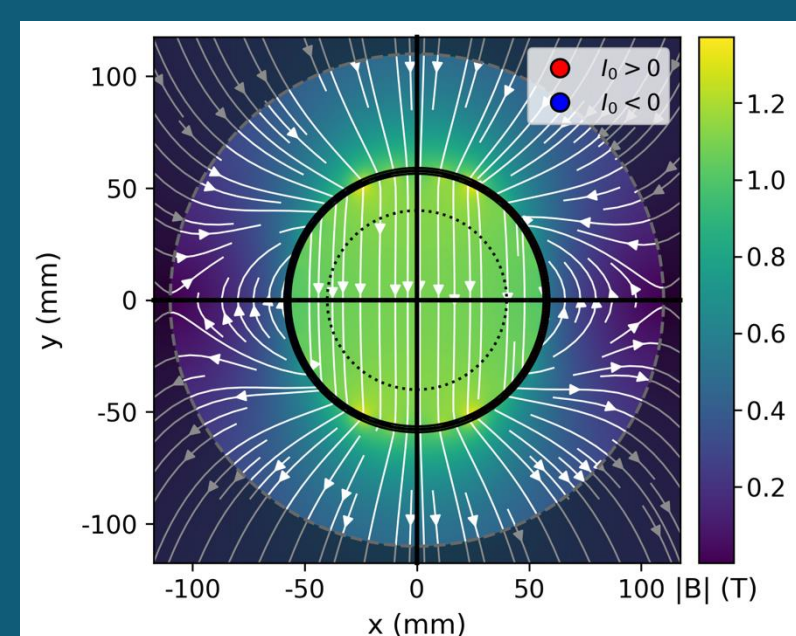
- MADEO also builds upon existing practices, and allows magnet designers to perform complex calculations and visualizations all within the same application, like above, where we see the field along the Z=0 line and a 3D model, both for a 12-layer magnet design
- It also allows magnet designers to export their final, optimized product to a winding file, allowing designers to move more efficiently from design to actual production

Methods: Modeling & Optimization:

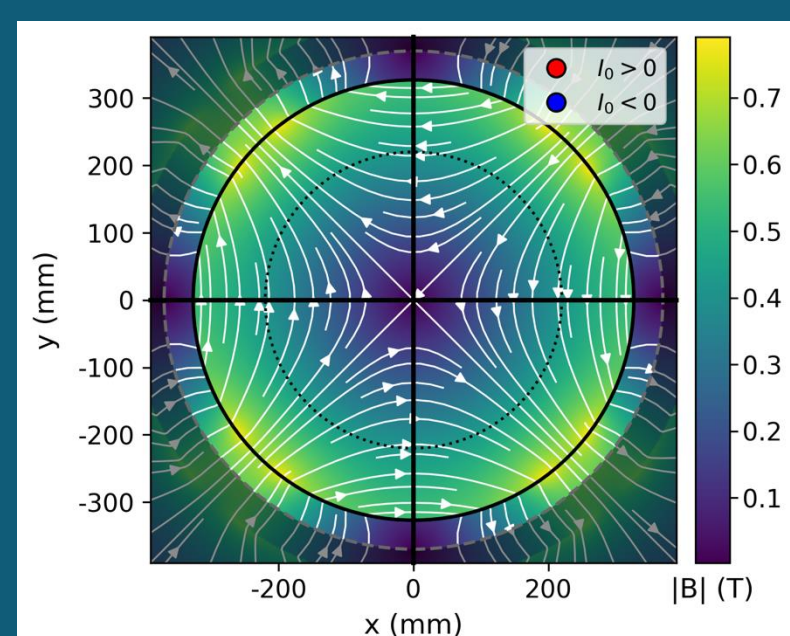


(Fig. 4: The GUI interface for MADEO that a user would see on start-up)

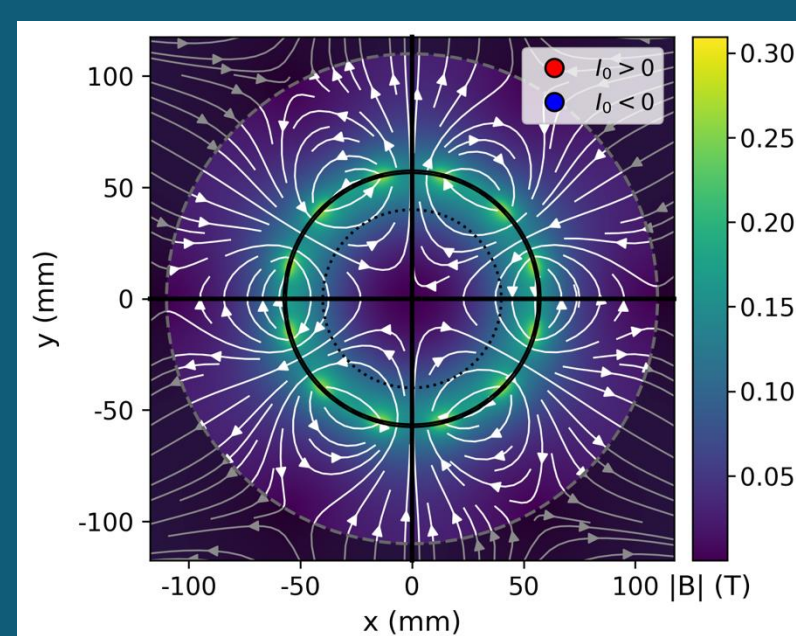
- To use MADEO, the user gives a collection of settings, from which an initial magnet design can be modeled. This is done via config files or through the user manually inputting each setting themselves.
- These direct-wind magnets are modeled by plotting conductors in an arrangement provided by the user settings. Calculations are then done, of both the individual harmonics and the total magnetic field through the region
- The user can then run this preliminary design through an optimizer, that will then vary geometric parameters for each block in each layer until a configuration with an optimal field quality is found
- In order to optimize these parameters, we use the iminuit Python packages, which allows us to find these minimal values very quickly



(Fig. 5a: A dipole magnet modeled with MADEO)



(Fig. 5b: A quadrupole magnet modeled with MADEO)



(Fig. 5c: A sextupole magnet modeled with MADEO)

Future Upgrades:

The program structure has been developed such that it can be upgraded to include several additional capabilities. This includes the possibility to vary the conductor count within each block. This new capability will allow many optimizations to be carried out in a single run and enhance the search envelope. In addition, the current version offers limited pre-processing and post-processing. The structure exists to read input files created for the old Fortran code to further process and optimize those designs. Similarly, the basic formulation and analytical tools have been developed and incorporated in the code to enhance the post-processing capabilities. Finally, the program can be modified to generate output files for other programs where non-linear yoke iron calculations can be performed.

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Conclusion:

Through this project, we were able to successfully develop MADEO: a more modern, Python-based program for modeling and optimizing direct wind magnets with an intuitive and easy to use GUI. In addition to developing an algorithm to model and optimize the conductor configurations and the fields they produce, we also laid the groundwork for a more user-friendly replacement to the existing legacy Fortran code currently used to optimize magnet designs. Further adoption of MADEO will allow magnet designers at Brookhaven National Lab to be able to simulate magnets much quicker and not have to worry about editing any sort of files or changing any code themselves. Most importantly, it will give the magnet designers the ability to see their designs and optimization in real time.

References:

- [1] Brookhaven National Laboratory. "Initial Winding of BEPCII Magnets." *BNL*, <https://www.bnl.gov/magnets/bepcii/initial-winding.php>.
- [2] Gupta, R. *Improving the Design and Analysis of Superconducting Magnets for Particle Accelerators*. Brookhaven National Laboratory, 1996, <https://wpw.bnl.gov/rgupta/wp-content/uploads/sites/9/2023/02/thesisrg.pdf>.